

ProxyFair: Fairness-Aware Causal Discovery via Proxy-Excess Regularization

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Abstract—Causal discovery infers graphical structure from observational data. In fairness-sensitive settings, this poses a risk: when the data reflect historical discrimination, a statistically plausible causal graph may still encode discriminatory pathways as part of the causal structure. Existing fairness interventions for causal structure learning face a dilemma: hard edge constraints can express categorical prohibitions but not degrees of proxy involvement, whereas directed fairness penalties provide a graded response but can break the score equivalence that equivalence-class search requires. We introduce *ProxyFair*, a fairness-aware extension of Greedy Equivalence Search (GES) that combines hard normative constraints with a continuous edge-level proxy penalty based on the *proxy-excess fraction*, which measures how much of an edge’s empirical support is reduced after conditioning on protected attributes. To preserve score equivalence, the method uses a symmetric penalty during completed partially directed acyclic graph (CPDAG) search, followed by directed fair directed-acyclic-graph (DAG) extraction within the learned equivalence class. We also study *constraint displacement*, in which suppressing one protected attribute’s structural influence can redistribute discriminatory signal through correlated attributes or proxies. Synthetic experiments show that ProxyFair suppresses planted proxy-mediated effects while preserving legitimate causal paths, and that joint multi-attribute penalization reduces displacement. Across four real-world fairness benchmarks, ProxyFair with only the soft proxy penalty consistently reduces total causal effects with moderate utility loss, while ProxyFair with both hard edge constraints and the soft proxy penalty achieves the strongest total-effect reductions on three of four datasets at the cost of larger fairness–utility trade-offs.

Index Terms—causal discovery, algorithmic fairness, proxy discrimination, structure learning, equivalence class search

I. INTRODUCTION

Causal discovery learns graphical structure from data, and the learned graph directly shapes downstream decisions. It determines which variables are treated as admissible predictors, which pathways are audited for discrimination, and which adjustment sets are used for causal estimation. This is particularly serious in fairness-sensitive domains: when observational data reflect historical discrimination, a statistically faithful graph may inherit that discrimination in its structure.

Structural Causal Models (SCMs) represent causal relationships as directed acyclic graphs (DAGs), but in observational settings causal discovery typically recovers only a Markov equivalence class (MEC) rather than a unique DAG [1]–[4]. Graphs in the same MEC agree on observed conditional independences but may orient some edges differently. In

fairness-sensitive settings, these alternative orientations can imply different pathways from protected attributes to outcomes and, therefore, different fairness conclusions.

Existing fairness interventions leave two practical issues unresolved. Hard constraints can forbid or require specific edges, but they are categorical and cannot express partial proxy involvement. Directed fairness penalties provide a graded response, but they assign different scores to different orientations of the same adjacency and therefore can break the score equivalence required by Greedy Equivalence Search (GES).

We introduce *ProxyFair*, a fairness-aware extension of GES that combines hard normative constraints with a continuous edge-level proxy penalty. The key idea is to penalize proxy-sensitive adjacencies symmetrically during CPDAG search, and to apply directed fairness only after the equivalence class has been learned. We also identify *constraint displacement*, where reducing one protected attribute’s structural influence can shift discriminatory signal through correlated protected attributes or proxy pathways.

Contributions.

- 1) We formulate fairness-aware causal discovery as a constrained structure-learning problem with proxy-excess regularization over equivalence classes.
- 2) We introduce the **proxy-excess fraction**, an edge-level statistic measuring the proportional drop in pairwise association after conditioning on protected attributes.
- 3) We develop ProxyFair, a two-phase extension of GES that uses a symmetric proxy penalty for score-equivalent CPDAG search and a directed criterion for fair DAG extraction.
- 4) We identify **constraint displacement** and provide empirical evidence that joint multi-attribute penalization can reduce this failure mode while preserving an explicit fairness–utility trade-off.

Code and supporting material: <https://github.com/VivianJiang-GingerRose/proxyfair-ges>.

II. BACKGROUND AND RELATED WORK

A. Causal Discovery and Greedy Equivalence Search

A standard framework for causal modeling is the SCM [1], which encodes relationships among variables as a DAG $\mathcal{G}$ over

variables $\mathbf{V}$, where an edge $V_j \rightarrow V_i$ asserts that V_j is a direct cause of V_i . In most real-world settings, this graph is unknown and must be inferred from observational data—a task known as causal discovery [2], [3].

Under the Causal Markov and Faithfulness assumptions, the conditional independence structure of the data identifies at best a MEC: the set of all DAGs entailing the same d-separation relations [4], [5]. The MEC is represented as a *Completed Partially Directed Acyclic Graph* (CPDAG), with directed edges where the orientation is the same for all DAGs in the set, and undirected edges where it is not.

Greedy Equivalence Search (GES) [6] recovers this CPDAG by maximizing a decomposable scoring function—typically the Bayesian Information Criterion (BIC) [7] or the Bayesian Dirichlet equivalent uniform (BDeu) score [8]—over the space of equivalence classes. This requires *score equivalence*: any two DAGs in the same MEC must receive identical scores [8], [9]. Integrating fairness penalties while preserving this property is the central technical challenge we address in Section IV.

B. Causal Fairness

Causal approaches to fairness decompose the total effect (TE) of a protected attribute S on outcome Y into the *natural direct effect* (NDE) and the *natural indirect effect* [10]. *Path-specific effects* (PSEs) refine this decomposition to individual pathways [11], though not all PSEs are identifiable from observational data. Kilbertus et al. [12] formalize the distinction between *resolving* variables, whose influence on Y is legitimate, and *proxy* variables, whose influence operates through group membership—a taxonomy that directly informs our proxy-excess fraction (Section IV-B). The legal dimension of proxy discrimination is discussed in Weerts et al. [13] and Prince & Schwarcz [14].

ProxyFair operates at the group level: it penalizes structural pathways that transmit S 's influence to Y across the population rather than requiring individual-level counterfactual reasoning [15]. For a comprehensive taxonomy of causal fairness notions, see Plečko & Bareinboim [16] and Makhoul et al. [17].

C. Related Work

Fair prediction and representation methods typically intervene on features, representations, or classifiers after a prediction task has been specified. Instead, ProxyFair targets the structure-learning stage, where the causal graph itself becomes the object used for downstream auditing and estimation.

Fair causal discovery methods most closely related to ProxyFair fall into two families.

a) *Constraint-based approaches*: Background knowledge can be incorporated into causal discovery as forbidden or required edges, with implications propagated through orientation rules such as Meek's rules [18]. Fairness constraints have similarly been used to restrict the set of admissible causal structures. Nabi and Shpitser [19] restrict structure learning to graphs satisfying path-specific fairness criteria, while Chiappa [20] derives counterfactual fairness constraints over partially oriented graphs. PC-Fairness [21] provides a

unified measurement framework for causality-based fairness notions, and Bonchi et al. [22] use PC (Peter–Clark)-style discovery to expose the causal structure of discrimination. These methods show that structural constraints can encode fairness requirements, but such constraints are inherently categorical: a pathway is either forbidden or allowed. Partial proxies, whose predictive value may mix legitimate information with protected-attribute signals, require a graded response that hard constraints alone cannot provide.

b) *Score-based approaches*: Score-based methods instead modify the discovery objective so that fairness-sensitive structures receive lower scores. FaGGM (Fairness-Aware Generative Graphical Models) [23] is the closest score-based fairness-aware Bayesian network (BN) learner, but it addresses a related rather than identical setting. It learns DAGs for fair synthetic data generation and evaluates downstream classifiers trained on that data. ProxyFair instead targets equivalence-class causal discovery and evaluates causal-effect diagnostics directly. Other work measures or analyzes fairness on already-learned graphs [21], [24], or assumes a known causal graph when regularizing proxy influence. ProxyFair addresses a complementary gap: it integrates an edge-level graded proxy penalty directly into GES-style equivalence-class search while preserving score equivalence and supporting causal-effect and displacement diagnostics.

III. PROBLEM FORMULATION

Let $\mathbf{D}$ be a dataset of N observations over variables $\mathbf{V} = \{V_1, \dots, V_p\}$. Let $\mathbf{S} \subset \mathbf{V}$ denote the protected attributes and $Y \in \mathbf{V}$ the outcome. A normative framework F encodes domain and fairness commitments as a constraint set $\mathcal{C}_F = (\mathcal{W}, \mathcal{B}_F)$, where $\mathcal{W} \subseteq \mathbf{V} \times \mathbf{V}$ is a whitelist of required directed edges and $\mathcal{B}_F \subseteq \mathbf{V} \times \mathbf{V}$ is a blacklist of forbidden directed edges. $E(\mathcal{G})$ denotes the edge set of $\mathcal{G}$.

We write $\Omega(\mathcal{C}_F)$ for the set of DAGs that satisfy these constraints:

$$\Omega(\mathcal{C}_F) = \{\mathcal{G} : \mathcal{G} \text{ is a DAG, } \mathcal{W} \subseteq E(\mathcal{G}), E(\mathcal{G}) \cap \mathcal{B}_F = \emptyset\}. \quad (1)$$

Problem 1 (Fair Causal Discovery). *Given $\mathbf{D}$, protected attributes $\mathbf{S}$, outcome Y , and constraints $\mathcal{C}_F$, learn a graph $\mathcal{G}^* \in \Omega(\mathcal{C}_F)$ that balances statistical fit to $\mathbf{D}$ with reduced protected-attribute influence on Y . Concretely, for a base decomposable score $\text{Score}_{\text{base}}$, we seek graphs of the form*

$$\mathcal{G}^* = \arg \max_{\mathcal{G} \in \Omega(\mathcal{C}_F)} [\text{Score}_{\text{base}}(\mathcal{G}; \mathbf{D}) - \lambda_f N \Psi(\mathcal{G}; \mathbf{S}, Y)], \quad (2)$$

where $\Psi(\mathcal{G}; \mathbf{S}, Y)$ measures the fairness-sensitive influence of protected attributes on the outcome through the learned structure, and $\lambda_f \geq 0$ controls the fairness–fit trade-off. The factor N places the penalty on the same scale as the log-likelihood; see Section IV-C.

This formulation separates two requirements: *constraint satisfaction*, which enforces categorical domain or normative prohibitions through $\mathcal{C}_F$, and *proxy regularization*, which discourages statistically supported but fairness-sensitive edges

Algorithm 1 ProxyFair: Fairness-Regularized Causal Discovery**Input:** Data $\mathbf{D}$, protected attributes $\mathbf{S}$, outcome Y , normative framework F **Output:** Fair causal graph $\mathcal{G}^*$ **Pre-computation**

- 1: Compute proxy array π^Σ and outcome relevance ρ_{adj} from $\mathbf{D}$ (Def. 1–2)

- 2: Elicit hard constraints $\mathcal{C}_F = (\mathcal{W}, \mathcal{B}_F)$ from F

Phase 1: Structure search

- 3: Run constrained GES using $\text{Score}_{\text{Proxy}}$ (Eq. 8)

- 4: Obtain CPDAG $\mathcal{C}^*$

Phase 2: Orientation selection

- 5: Select $\mathcal{G}^* \leftarrow \arg \min_{\mathcal{G} \in [\mathcal{C}^*] \cap \Omega(\mathcal{C}_F)} \Psi_{\text{Total}}(\mathcal{G})$ (Eq. 11)

- 6: **return** $\mathcal{G}^*$

within the feasible region. Because GES searches over equivalence classes, a directional fairness penalty cannot be inserted into the score directly. ProxyFair therefore implements Eq. (2) in two stages: Phase 1 uses a symmetric proxy penalty during CPDAG search, while Phase 2 applies a directed fairness criterion within the learned equivalence class. Algorithm 1 summarizes the pipeline.

IV. THE PROXYFAIR METHOD

ProxyFair combines hard normative constraints with a continuous proxy penalty in a two-phase procedure. The core difficulty is that fairness is directional: whether an edge is fairness-sensitive depends on which variable is the parent and which is the child. A directed penalty—one that penalizes $V_j \rightarrow V_i$ differently from $V_i \rightarrow V_j$ —may assign different scores to DAGs within the same MEC, thereby breaking the score-equivalence property on which GES relies. ProxyFair addresses this by using a symmetric penalty during CPDAG search (Phase 1) and applying directed fairness preferences only after the equivalence class has been recovered (Phase 2).

A. Hard Normative Constraints

1) *Constraint Types*: ProxyFair separates domain constraints from normative constraints. *Type A constraints* encode framework-independent structural restrictions, such as temporal ordering, self-causation, and ontological impossibilities. For example, an outcome cannot cause immutable or temporally prior attributes such as age, race, gender, or birth country. *Type B constraints* encode framework-specific prohibitions: Anti-Classification forbids direct protected-attribute-to-outcome edges, whereas stronger anti-proxy frameworks may also blacklist proxy-to-outcome edges. ProxyFair is agnostic to the chosen framework and accepts the resulting constraint set $\mathcal{C}_F = (\mathcal{W}, \mathcal{B}_F^A \cup \mathcal{B}_F^B)$ as input.

Hard constraints integrate into GES as search-space restrictions. Blacklisted edges in $\mathcal{B}_F$ are excluded from every forward-phase edge-addition step. Whitelisted edges in $\mathcal{W}$ are initialized in the graph before search begins and protected from removal. Because GES evaluates operations via score *deltas*, a whitelisted edge—present in both configurations of any

candidate operation on other edges—has its penalty contribution cancel in the delta, requiring no special treatment in the scoring phase.

B. Proxy-Excess Fraction

A natural starting point for fairness-aware scoring is to penalize edges according to *node-level* proxy strength, such as the marginal association between a variable and a protected attribute. This is insufficiently specific: a variable Z may be correlated with protected attribute S while still predicting outcome Y through associations that are largely independent of S . Penalizing every edge incident to Z therefore risks suppressing legitimate predictive pathways alongside fairness-sensitive ones.

ProxyFair instead uses an *edge-level* regularizer. For a candidate adjacency $\{V_j, V_i\}$, the regularizer compares the marginal association between V_j and V_i with the association that remains after conditioning on a protected attribute.

Definition 1 (Proxy-Excess Fraction). *For variables V_j, V_i and protected attribute $S_k \in \mathbf{S}$, define the proxy-excess fraction:*

$$\pi(V_j, V_i, S_k) = \begin{cases} \max\left(0, \frac{r_{V_j V_i}^2 - r_{V_j V_i \cdot S_k}^2}{r_{V_j V_i}^2}\right) & \text{if } r_{V_j V_i}^2 > \tau \\ 0 & \text{otherwise,} \end{cases} \quad (3)$$

where $\tau = c/(N-2)$ with $c = 5$.

Here $r_{V_j V_i}^2$ denotes the pairwise association between V_j and V_i , and $r_{V_j V_i \cdot S_k}^2$ its conditional analogue after controlling for S_k . In our experiments, all variables are discretized before structure learning, so both quantities are computed using normalized mutual information and conditional normalized mutual information, respectively. For fully continuous settings, squared Pearson correlation and squared partial correlation could be used analogously, since the symmetry required by Proposition 1 would still hold.

Intuitively, $\pi(V_j, V_i, S_k)$ measures the proportional drop in the empirical support for adjacency $\{V_j, V_i\}$ after conditioning on S_k . The threshold τ prevents unstable fractions when the marginal association is close to the sampling-noise floor.

We aggregate proxy-excess fractions additively across protected attributes:

$$\pi^\Sigma(V_j, V_i) = \sum_{S_k \in \mathbf{S}} \pi(V_j, V_i, S_k). \quad (4)$$

We use $\pi^\Sigma(V_j, V_i)$ for the scalar proxy-excess value of a variable pair, and π^Σ for the full precomputed array. Because individual π values need not sum to one, π^Σ is an unnormalized cumulative score whose scale is controlled by λ_f . Additive aggregation treats each protected attribute as independently relevant for fairness scrutiny, while accepting the possibility of double-counting when protected attributes are correlated; Appendix B-A discusses alternatives.

Because π is computed before the graph is known, it is a regularization signal rather than a causal estimand. A high value indicates that the association supporting edge $\{V_j, V_i\}$

depends strongly on protected-attribute information, but it does not by itself distinguish mediation from confounding (see Appendix B-B).

Symmetry. Score equivalence requires the penalty on edge $\{V_j, V_i\}$ to be independent of orientation. Mutual information and conditional mutual information are symmetric in V_i and V_j ; the same holds for squared correlation and squared partial correlation if continuous variables are retained. Hence $\pi(V_j, V_i, S_k) = \pi(V_i, V_j, S_k)$, and therefore $\pi^\Sigma(V_j, V_i) = \pi^\Sigma(V_i, V_j)$.

C. Phase 1: Fairness-Regularized Search

During the GES forward and backward phases, we augment the base score with a symmetric fairness penalty anchored by the proxy-excess fraction. A second precomputed quantity focuses the penalty on edges that matter for the outcome.

Definition 2 (Outcome Relevance). *The outcome relevance of variable V_i is:*

$$\rho_{\text{adj}}(V_i) = \begin{cases} 1 & \text{if } V_i = Y \\ \rho(V_i, Y) & \text{otherwise} \end{cases} \quad (5)$$

where $\rho(V_i, Y)$ is the marginal association between V_i and Y .

This term prevents ProxyFair from penalizing edges that are structurally remote from the outcome as heavily as edges near Y . Phase 2 replaces this marginal relevance score with a graph-aware alternative.

The directed penalty for an edge $V_j \rightarrow V_i$ is:

$$\psi(V_j \rightarrow V_i) = \pi^\Sigma(V_j, V_i) \cdot \rho_{\text{adj}}(V_i) \quad (6)$$

This is asymmetric in general: $\rho_{\text{adj}}(V_i)$ depends on which node plays the role of child. To preserve score equivalence, we symmetrize by averaging over both orientations:

$$\Psi_{\text{sym}}(V_i, V_j) = \frac{1}{2} (\psi(V_i \rightarrow V_j) + \psi(V_j \rightarrow V_i)) \quad (7)$$

We use the mean rather than min or max: min can be too lenient when only one orientation is fairness-relevant, whereas max can over-penalize edges whose unfairness depends on an unlikely orientation.

Let $S_b(V_i) = \text{Score}_{\text{base}}(V_i, \text{Pa}(V_i))$ denote any decomposable, score-equivalent base score (e.g., BIC or BDeu). The fairness-regularized local score is:

$$S_{\text{Proxy}}(V_i) = S_b(V_i) - \lambda_f N \sum_{V_j \in \text{Pa}(V_i)} \Psi_{\text{sym}}(V_i, V_j) \quad (8)$$

The factor N scales the fairness penalty to match the base score's log-likelihood term. Without it, the penalty would vanish relative to statistical fit as sample size grows, allowing discriminatory but well-supported edges to re-enter. Since π^Σ converges to a population quantity, λ_f remains the main control on the fairness-utility trade-off; Appendix B-C gives the full argument.

Because Ψ_{sym} is symmetric and precomputed, the penalty GES evaluates for edge $\{V_i, V_j\}$ is the same regardless of how the resulting CPDAG orients that edge, ensuring clean integration into equivalence-class search.

D. Phase 2: Fair DAG Extraction

GES returns a CPDAG $\mathcal{C}^*$: an equivalence class of DAGs that are statistically indistinguishable but may differ in fairness consequences, since different orientations can place a proxy variable on or off a causal pathway to Y .

Definition 3 (Directed Fairness Penalty). *For a fully oriented DAG $\mathcal{G}$, define:*

$$\Psi_{\text{Total}}(\mathcal{G}) = \sum_{V_i \in \mathbf{V}} \rho_{\mathcal{G}}(V_i) \cdot \sum_{V_j \in \text{Pa}_{\mathcal{G}}(V_i)} \pi^\Sigma(V_j, V_i) \quad (9)$$

where the graph-aware outcome relevance is:

$$\rho_{\mathcal{G}}(V_i) = \begin{cases} 1 & \text{if } V_i = Y \text{ or } V_i \in \text{Pa}_{\mathcal{G}}(Y) \\ \omega & \text{if } V_i \in \text{CP}_{\mathcal{G}}(Y) \setminus \text{Pa}_{\mathcal{G}}(Y) \\ \beta \cdot \gamma(V_i, Y) & \text{otherwise} \end{cases} \quad (10)$$

Here $\text{Pa}_{\mathcal{G}}(Y)$ denotes the parents of Y , and $\text{CP}_{\mathcal{G}}(Y)$ denotes co-parents, i.e., nodes sharing a child with Y but not directly connected to it. The fallback term $\gamma(V_i, Y)$ is the same marginal outcome association used in Definition 2. The weights $\omega, \beta \in [0, 1]$ control the relative penalty on co-parent and remote edges, respectively. We set $\omega = 1$ throughout because co-parents enter the conditioning set for causal-effect estimation. We set $\beta = 0$ in all experiments, so edges outside Y 's structural neighborhood do not affect the reported Phase 2 selection.

In contrast to this directed criterion, the symmetric Phase 1 penalty depends only on unordered adjacencies and therefore preserves score equivalence (proof in Section IV-E).

Phase 2 selects the fairest constraint-feasible extension of the learned CPDAG:

$$\mathcal{G}^* = \arg \min_{\mathcal{G} \in [\mathcal{C}^*] \cap \Omega(\mathcal{C}_F)} \Psi_{\text{Total}}(\mathcal{G}). \quad (11)$$

Because DAGs in $[\mathcal{C}^*]$ share the same Phase 1 score, Phase 2 only minimizes the directed fairness criterion. The intersection with $\Omega(\mathcal{C}_F)$ prevents Phase 2 from selecting orientations that violate required or forbidden edges.

With $\beta = 0$, Ψ_{Total} depends only on Y 's structural neighborhood. Enumeration can be exponential in undirected edges but was feasible for our benchmarks.

E. Score Equivalence

The symmetric penalty is not merely a design heuristic; it provably preserves the score-equivalence property required by GES.

Proposition 1 (Phase 1 Score Equivalence). *The Phase 1 score $\text{Score}_{\text{Proxy}}$ is score-equivalent: all DAGs in the same MEC receive identical Phase 1 scores.*

Proof. The total ProxyFair penalty can be written as

$$\sum_{V_i} \sum_{V_j \in \text{Pa}_{\mathcal{G}}(V_i)} \Psi_{\text{sym}}(V_i, V_j).$$

Since $\Psi_{\text{sym}}(V_i, V_j) = \Psi_{\text{sym}}(V_j, V_i)$, this equals

$$\sum_{\{V_i, V_j\} \in \text{Skel}(\mathcal{G})} \Psi_{\text{sym}}(V_i, V_j).$$

The penalty therefore depends only on which adjacencies are present in $\mathcal{G}$, not on how those adjacencies are oriented. Since all DAGs in the same MEC share the same skeleton, the ProxyFair penalty is invariant within the MEC. The base score is score-equivalent by assumption, so the sum of the base score and the ProxyFair penalty is also score-equivalent. $\square$

This result ensures that GES can optimize the fairness-regularized objective over equivalence classes without reducing the search to DAG space. The resulting optimum should be interpreted as optimal for the regularized objective, not as a consistent estimator of the unregularized data-generating DAG.

V. EXPERIMENTS

We evaluate ProxyFair on synthetic data with known ground truth, as well as on four real-world fairness benchmarks.

A. Setup

1) *Datasets: Synthetic.* We use two linear Gaussian structural equation models (SEMs), discretized into five equal-frequency bins. The *single-attribute* graph is $S \rightarrow V \rightarrow Y$ with $Z \rightarrow Y$, where $S \perp Z$ and $V \perp Z$ by construction: the path $S \rightarrow V \rightarrow Y$ is discriminatory while $Z \rightarrow Y$ is legitimate. The *multi-attribute* graph adds a second protected attribute: $C \rightarrow S_1$, $C \rightarrow S_2$, $S_1 \rightarrow V_1 \rightarrow Y$, $S_2 \rightarrow V_2 \rightarrow Y$, and $Z \rightarrow Y$, with an observed common cause C inducing $\text{corr}(S_1, S_2) \approx 0.5$. Unless otherwise stated, we use $N = 5,000$ observations and 50 random seeds. For the single-attribute setting, proxy strength $\phi^*(V, S)$ (Cramér’s V) is fixed at 0.5.

Real-world. We use Law School [25] ($N=20,798$; race, gender $\rightarrow$ bar passage), COMPAS [26] ($N=6,167$; race, gender $\rightarrow$ recidivism), Dutch Census [27] ($N=60,420$; gender, birth country, citizenship $\rightarrow$ occupation level), and Bank Marketing [28] ($N=45,211$; age, marital status $\rightarrow$ deposit subscription). Because the real-world benchmarks contain mixed-type attributes, all variables are discretized before structure learning. Preprocessing choices, excluded variables, and final preprocessed datasets are included in the anonymous repository.

2) *Constraint Elicitation:* ProxyFair treats the set of normative constraints $\mathcal{C}_F$ as input. In our experiments, we instantiate it by classifying variables as protected, proxy, legitimate, or outcome using a multi-provider large language model (LLM) protocol with majority voting, followed by deterministic rule application. The LLMs are used only for role elicitation; they do not infer causal structure, estimate effects, or modify the score. Any source of variable roles, including domain experts, regulatory guidance, prior audits, or dataset documentation, can instantiate the same pipeline. Protected-attribute validation is reported in Appendix A.

3) *Baselines and Ablations:* We compare methods that isolate whether fairness enters through post-search orientation, hard constraints, soft scoring, or post-hoc pruning. All GES searches use BDeu scoring [8]. **GES** is vanilla GES with no fairness intervention. **GES+Phase2** applies Phase 2 fair orientation selection to the vanilla GES CPDAG, testing

whether fair DAG extraction alone is sufficient without fairness-aware search. **GES+Hard** adds hard normative constraints $\mathcal{C}_F$ during GES but uses no soft proxy penalty. **PF-Soft** uses Phase 1 fairness-regularized scoring followed by Phase 2 orientation selection, but no hard constraints. **PF-Hard+Soft** is the full ProxyFair pipeline, combining hard constraints, Phase 1 fairness-regularized scoring, and Phase 2 orientation selection.

We also include two diagnostic baselines. **PostHocPruning** runs vanilla GES and then removes high-proxy adjacencies using the same proxy statistics as ProxyFair, testing whether post-search pruning can substitute for fairness-aware search. Only in the synthetic single-attribute experiment does GES+Hard blacklist $V \rightarrow Y$ conditionally on $\phi^*(V, S) > 0.5$; at the boundary $\phi^* = 0.5$ the rule does not activate, so only a graded penalty can respond.

To our knowledge, existing methods do not directly integrate graded fairness regularization into equivalence-class search while preserving GES score equivalence. Prior approaches either apply directed penalties outside equivalence-class search [23], measure fairness on already-learned structures [21], [24], or enforce categorical prohibitions without graded proxy penalties [19], [20]. We therefore evaluate against ablations of our own pipeline, isolating each component under the same equivalence-class search framework.

4) *Metrics:* We report three families of metrics.

Causal fairness.

Causal fairness. We report absolute total effect $|\text{TE}|$, natural direct effect $|\text{NDE}|$, and path-specific effect $|\text{PSE}|$ from protected attribute S to outcome Y . For multi-attribute datasets, metrics are computed separately for each protected attribute and reported as unweighted means of absolute values. Given a learned DAG, we fit local predictive models for each node conditional on its parents and estimate effects using graph-defined interventions. PSE is reported only when the relevant proxy-mediated intervention is well-defined; otherwise it is marked N/A. These estimates inherit any bias from structure-learning error, model misspecification, or unobserved confounding.

Group fairness. Statistical parity difference (SPD) measures the gap in positive-outcome rates across protected groups; equalized odds difference (EO) measures differences in true-positive and false-positive rates across groups. For both, lower is better.

Predictive performance. We report the area under the receiver operating characteristic curve (AUROC) of a downstream XGBoost classifier trained on the learned DAG’s Markov blanket of Y ; higher AUROC is better. For each real-world dataset, we use the same fixed stratified 80/20 train–test split across all methods, with five-fold cross-validation on the training split.

For datasets with multiple protected attributes, the main tables report unweighted means across attributes, with full per-attribute results included in the anonymized repository. For methods that return a CPDAG without selecting a unique DAG, we report mean $\pm$ standard deviation over CPDAG-consistent DAG extensions. This captures equivalence-class

TABLE I
SYNTHETIC RESULTS AT $\phi^* = 0.5$, $N = 5,000$, OVER 50 SEEDS. MEAN $\pm$ SD REPORTED.

Algorithm (λ_f)	TE $\downarrow$	PSE $\downarrow$	NDE $\downarrow$	SPD $\downarrow$	EO $\downarrow$	AUROC $\uparrow$	Modal graph (freq.)
GES (—)	0.401 $\pm$ 0.101	0.322 $\pm$ 0.059	0.082 $\pm$ 0.083	0.539 $\pm$ 0.148	0.391 $\pm$ 0.158	0.872 $\pm$ 0.029	$S \rightarrow V, V \rightarrow Y, Z \rightarrow Y$ (54%)
GES+Hard (—)	0.360 $\pm$ 0.071	0.359 $\pm$ 0.071	0.008 $\pm$ 0.005	0.510 $\pm$ 0.129	0.374 $\pm$ 0.135	0.870 $\pm$ 0.027	$S \rightarrow V, V \rightarrow Y, Z \rightarrow Y$ (100%)
PF-Soft (0.5)	0.253 $\pm$ 0.197	0.254 $\pm$ 0.195	0.008 $\pm$ 0.005	0.281 $\pm$ 0.281	0.213 $\pm$ 0.208	0.816 $\pm$ 0.081	$S \rightarrow V, V \rightarrow Y, Z \rightarrow Y$ (62%)
PF-Soft (1.0)	0.007 $\pm$ 0.006	0.009 $\pm$ 0.006	0.008 $\pm$ 0.006	0.026 $\pm$ 0.008	0.043 $\pm$ 0.009	0.695 $\pm$ 0.066	$S \rightarrow V, Z \rightarrow Y$ (84%)
PF-Soft (2.0)	0.008 $\pm$ 0.006	0.009 $\pm$ 0.005	0.008 $\pm$ 0.006	0.027 $\pm$ 0.008	0.040 $\pm$ 0.010	0.696 $\pm$ 0.061	$Z \rightarrow Y$ (78%)

variation rather than data-resampling uncertainty. For methods that select a single DAG, including Phase 2-based methods, we report the selected-DAG value.

5) *Hyperparameters*: For synthetic experiments, we report $\lambda_f \in \{0.5, 1.0, 2.0\}$ to show the fairness–utility trade-off. For real-world datasets, λ_f controls the weight of proxy-excess regularization relative to statistical fit. We recommend sweeping λ_f over a log-spaced grid and selecting an interior point from the fairness–AUROC Pareto frontier according to domain-specific utility tolerance. We illustrate this on COMPAS with a dense sweep over $\lambda_f \in [0.05, 20]$ (reporting $\lambda_f=1.9$) and select interior points from smaller sweeps on the remaining datasets: $\lambda_f = 0.1$ for Law and $\lambda_f = 0.5$ for Dutch and Bank.

a) *Reproducibility and compute.*: The implementation and supplementary per-attribute results are available in the anonymous repository. Synthetic experiments use $N=5,000$ and 50 random seeds. Real-world results use a single fixed train–evaluation protocol per dataset; for CPDAG-based entries, mean $\pm$ SD reflects equivalence-class variation. All experiments run on a single CPU without GPU acceleration. Representative real-world runs take approximately 1–24 hours, depending on how many consistent DAGs are represented by the learned CPDAG.

B. Synthetic Evaluation

We first evaluate under known ground truth, testing proxy suppression and constraint displacement on the synthetic graphs described in Section V-A.

1) *Single-Attribute Results*: Table I presents the single-attribute proxy-suppression results.

Threshold blindness. Both GES and GES+Hard retain $V \rightarrow Y$ in all 50 seeds. At the boundary $\phi^* = 0.5$, the categorical proxy rule does not intervene.

Graded proxy suppression. ProxyFair exhibits a clear fairness–utility trade-off as λ_f increases. At $\lambda_f = 1.0$, the proxy edge $V \rightarrow Y$ is removed in most seeds, the modal graph $S \rightarrow V, Z \rightarrow Y$ appears in 84% of runs, and |TE| drops to 0.007 ± 0.006 . Increasing to $\lambda_f = 2.0$ yields little additional fairness gain but removes more upstream structure, indicating saturation of the useful penalty range.

Legitimate paths are preserved. In the edge-frequency results, $Z \rightarrow Y$ is retained in all seeds across the reported λ_f values. The AUROC reduction is therefore attributable to the removal of the planted proxy path rather than to the loss of the legitimate causal path.

TABLE II
MULTI-ATTRIBUTE DISPLACEMENT UNDER KNOWN GROUND TRUTH.
 $N=5,000$, 50 SEEDS, $\lambda_f=1.0$. MEAN $\pm$ SD REPORTED.

Penalty set	PSE _{S₁} $\downarrow$	PSE _{S₂} $\downarrow$	PSE _{total} $\downarrow$	Pr[$Z \rightarrow Y$] $\uparrow$
None	0.300 $\pm$ 0.007	0.300 $\pm$ 0.006	0.600 $\pm$ 0.007	1.00
{ S_1 }	0.000$\pm$0.000	0.367 $\pm$ 0.008	0.367 $\pm$ 0.008	1.00
{ S_2 }	0.367 $\pm$ 0.009	0.000$\pm$0.000	0.367 $\pm$ 0.009	1.00
{ S_1, S_2 }	0.000$\pm$0.000	0.000$\pm$0.000	0.000$\pm$0.000	1.00

2) *Multi-Attribute Results*: Table II shows constraint displacement directly.

Penalizing only S_1 removes the planted $S_1 \rightarrow V_1 \rightarrow Y$ path, but increases |PSE_{S₂}| from 0.300 to 0.367, a 22% amplification rather than mere persistence. Penalizing only S_2 produces the symmetric effect. This amplification arises because S_1 and S_2 share the observed common cause C and are therefore correlated. Once the penalized $S_1 \rightarrow V_1 \rightarrow Y$ route becomes less attractive, the score can recover part of the lost fit through the correlated but unpenalized $S_2 \rightarrow V_2 \rightarrow Y$ route. The intervention therefore redistributes discriminatory signal rather than eliminating it.

Joint penalization through additive π^Σ drives both planted proxy PSEs to zero while retaining $Z \rightarrow Y$ in every seed. Thus, when protected attributes are correlated, single-attribute mitigation can shift discriminatory influence through an unpenalized attribute, whereas additive multi-attribute penalization suppresses both paths.

C. Real-World Benchmarks

Table III presents results across all four datasets.

a) *Soft proxy regularization reduces total causal effects*: PF-Soft reduces |TE| relative to GES on all four datasets: 83% on Law (0.067 $\rightarrow$ 0.012), 79% on Dutch (0.085 $\rightarrow$ 0.018), 74% on Bank (0.017 $\rightarrow$ 0.005), and 29% on COMPAS (0.023 $\rightarrow$ 0.016). The magnitude varies by dataset, likely reflecting differences in proxy structure. On COMPAS, where the TE reduction is smallest, the AUROC cost is also minimal (0.619 $\rightarrow$ 0.613).

b) *Hard and soft constraints are complementary*: PF-Hard+Soft achieves the lowest |TE| on three of four datasets and the best |PSE| on three of four, suggesting that categorical prohibitions and graded proxy penalties target different structural failures. The trade-off is larger AUROC loss on some

TABLE III

REAL-WORLD BENCHMARK RESULTS. PF DENOTES PROXYFAIR. LOWER IS BETTER FOR FAIRNESS METRICS; HIGHER IS BETTER FOR AUROC. BEST VALID VALUE PER DATASET AND METRIC IS BOLDED.

Data	Method	Causal fairness			Group fairness		Perf.
		TE ↓	PSE ↓	NDE ↓	SPD ↓	EO ↓	AUROC ↑
Law	GES	.0667 ± .0261	.0066 ± .0030	.0563 ± .0211	.0936 ± .0338	.2370 ± .0395	.6129 ± .0056
	GES+Phase2	.0205	.0106	.0237	.1190	.2434	.5998
	GES+Hard	.0170	.0040	.0137	.0227	.0416	.4879
	PostHocPruning	.0090 ± .0030	N/A [‡]	.0083 ± .0039	.0184 ± .0059	.0906 ± .0041	.5132 ± .0153
	PF-Soft	.0115	.0055	.0062	.0248	.0940	.5860
	PF-Hard+Soft	.0061	.0045	.0063	.0146	.0493	.5063
Dutch	GES	.0846 ± .0205	.0116 ± .0081	.0784 ± .0105	.1991 ± .0735	.1834 ± .0790	.8338 ± .0200
	GES+Phase2	.0395	N/A [†]	.0525	.2387	.1999	.8408
	GES+Hard	.0139	.0123	.0098	.1113	.1775	.7131
	PostHocPruning	.0353 ± .0096	.2060 ± .0655, [‡]	.0443 ± .0074	.2149 ± .0529	.1889 ± .0653	.8395 ± .0161
	PF-Soft	.0181	.0027	.0149	.0808	.1304	.8108
	PF-Hard+Soft	.0135	.0079	.0091	.1192	.2125	.7151
Bank	GES	.0174 ± .0039	.0022 ± .0008	.0152 ± .0047	.0324 ± .0040	.0992 ± .0392	.6417 ± .0230
	GES+Phase2	.0129	N/A [†]	.0111	.0597	.1866	.6527
	GES+Hard	.0069	.0086	.0051	.0255	.0919	.6091
	PostHocPruning	.0146 ± <.0001	N/A [‡]	.0133 ± .0002	.0426 ± .0011	.0699 ± .0208	.6683 ± .0187
	PF-Soft	.0045	.0062	.0107	.0153	.0928	.6148
	PF-Hard+Soft	.0031	.0012	.0024	.0131	.0276	.5194
COMPAS	GES	.0227 ± .0230	.0133 ± .0079	.0268 ± .0227	.0840 ± .0345	.2091 ± .0271	.6190 ± .0531
	GES+Phase2	.0575	.0212	.0442	.1525	.2118	.6089
	GES+Hard	.0826	.0109	.0646	.1743	.2431	.6772
	PostHocPruning	.0345 ± .0126	.0315 ± .0082	.0221 ± .0150	.1305 ± .0480	.2072 ± .0771	.6309 ± .0389
	PF-Soft	.0161	.0219	.0162	.0441	.1475	.6133
	PF-Hard+Soft	.0203	.0108	.0157	.1714	.2678	.6167

Notes. Mean ± SD reports equivalence-class or pruned-DAG variation, not resampling uncertainty; single values are selected-DAG results. Near-chance AUROC values indicate that the learned Markov blanket contains limited predictive signal after strong constraints and are interpreted as diagnostic frontier endpoints. [†] GES+Phase2 produces invalid or diagnostic protected-attribute orientations on Dutch and Bank, so these PSE entries are not treated as valid fairness improvements. [‡] PostHocPruning PSE is reported only when valid proxy-mediated PSE calculations remain.

datasets, especially Law, Dutch, and Bank. This appears partly driven by Type A temporal and ontological constraints, which remove causally incoherent edges that may still carry predictive signal.

c) Fairness gains can reach low-utility regimes: The strongest causal-effect reductions sometimes occur at the low-utility end of the trade-off curve. For example, PF-Hard+Soft attains the lowest |TE| on Law and Bank, but its AUROC falls close to the no-skill level (0.506 and 0.519, respectively). These configurations should be interpreted as diagnostic frontier endpoints rather than deployable choices. In practice, λ_f should be selected from the interior of the Pareto frontier, where causal effects are reduced while predictive performance remains meaningfully above chance.

d) Orientation and pruning alone are insufficient: GES+Phase2 tests whether fair orientation selection can repair a CPDAG learned without fairness-aware search. On COMPAS, it worsens both |TE| (0.023 → 0.058) and |PSE| (0.013 → 0.021). On Dutch and Bank, it produces invalid or diagnostic protected-attribute orientations, such as orienting temporally fixed attributes as children of the outcome. PostHocPruning shows less consistent behavior: it improves some metrics on Law but worsens causal fairness on COMPAS, yields a large valid PSE estimate on Dutch, and produces no valid post-pruning PSE calculations on Law and Bank. These results suggest that fairness should shape the equivalence class during search rather than only reorienting or pruning after discovery.

e) Causal and group fairness can diverge: PF-Soft is the most stable default across the real-world benchmarks, but the results also show that causal and group fairness do not always move together. On COMPAS, PF-Hard+Soft lowers NDE and PSE relative to PF-Soft but increases SPD from 0.044 to 0.171; on Dutch, a similar pattern appears for EO. The per-attribute COMPAS results show a related asymmetry: PF-Hard+Soft reduces race-related |TE| (0.0335 → 0.0076) but increases gender-related |TE| (0.0118 → 0.0331). One plausible interpretation is constraint displacement: when hard constraints remove pathways that carry both discriminatory and predictive signals, the remaining associations may be represented by correlated proxies. These results suggest that causal, group, and per-attribute diagnostics should be monitored together.

PF-Soft also increases orientation flexibility: its learned CPDAGs contain 21–135 consistent extensions, compared with 2–18 under baseline. Phase 2 uses this ambiguity to prefer orientations that avoid placing high- π^Σ variables as parents of Y . Figure 1 illustrates the COMPAS Pareto frontier across the λ_f sweep, showing the range of fairness–utility trade-offs obtained by PF-Soft.

VI. DISCUSSION AND LIMITATIONS

A. Constraint Displacement

Our experiments identify a failure mode we term *constraint displacement*: reducing one protected attribute’s structural

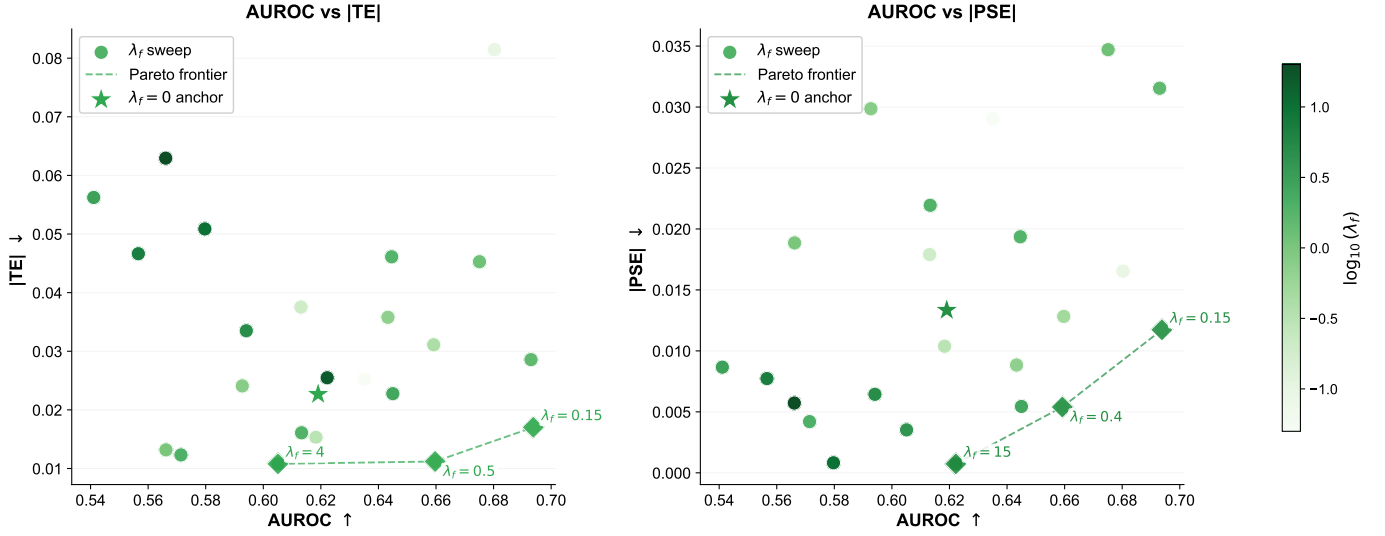


Fig. 1. Pareto frontier for PF-Soft on COMPAS across the λ_f sweep. Each point corresponds to one λ_f value, with color indicating $\log_{10}(\lambda_f)$. The same procedure can be used to select λ_f on new datasets.

influence can shift discriminatory signal through correlated attributes or proxies. Related redistribution effects have been observed in machine unlearning, where forgetting information about one demographic group can shift bias toward correlated groups rather than eliminating it [29]. Our synthetic multi-attribute results (Table II) isolate this effect under known ground truth, where additive π^Σ penalization removes it. The real-world COMPAS and Dutch results show consistent patterns, but should be interpreted cautiously because the ground-truth causal structures are unknown.

B. Limitations

ProxyFair inherits the assumptions of GES: causal sufficiency, the Markov condition, and faithfulness. The fairness penalty deliberately biases search away from statistically supported structures when those structures are proxy-sensitive, so downstream causal reasoning on penalized edges should be interpreted with care. When unobserved confounders correlate with protected attributes, the proxy-excess fraction may misattribute confounding to proxy mediation. Extending the framework to latent-variable methods such as Fast Causal Inference (FCI) is therefore an important direction for future work. Strong constraints or large fairness penalties can also produce graphs whose Markov blanket of Y has limited predictive value, so near-chance AUROC configurations should be treated as diagnostic frontier endpoints rather than deployable operating points.

Methodologically, π may underpenalize nonlinear proxy mechanisms, and additive aggregation in π^Σ does not model intersectional effects where combinations of attributes create discrimination beyond individual contributions [30]. Phase 2 enumeration can scale exponentially in the number of undirected edges [31], though it was feasible for our benchmarks. Finally, LLM-based elicitation reduces manual burden but

does not eliminate variability among model providers, so low-consensus classifications still require expert review.

VII. CONCLUSION

ProxyFair integrates fairness into GES-style causal discovery by combining hard normative constraints with a symmetric proxy-excess penalty that preserves score equivalence, followed by directed fair DAG extraction within the learned equivalence class. Synthetic experiments show that ProxyFair suppresses planted proxy-mediated pathways while retaining legitimate structure, and that joint multi-attribute penalization reduces constraint displacement. Across four real-world benchmarks, PF-Soft consistently reduces total protected-attribute effects with moderate utility loss, while PF-Hard+Soft reaches stronger causal reductions at the cost of larger fairness–utility trade-offs.

More broadly, these results suggest that proxy-excess regularization should shape the equivalence class during search, rather than only pruning or reorienting after discovery, while multi-attribute diagnostics are needed to detect whether fairness gains are displaced across protected attributes.

APPENDIX A LLM ELICITATION VALIDATION

Table IV compares LLM-elicited protected attributes against literature ground truth across all four datasets. T1 denotes strongly protected attributes with clear legal or empirical grounding; T2 denotes context-dependent attributes whose protected status depends on domain and deployment context. LLM consensus aligns with the literature ground truth for all T1 attributes. Our experiments use T1 attributes only; T2 entries are reported for transparency and represent candidates whose protected status is still a normative decision in the deployment context.

Dataset	Attribute	Literature Ground Truth (Priv. / Unpriv. Groups)	LLM Consensus (Tier: Priv. / Unpriv. Groups)
Law School	race	White / Non-White	T1: White / Non-White
	gender	Male / Female	T1: Male / Female
	family income	—	T2: Higher / Lower Income
	LSAT score	—	T2: Higher / Lower LSAT Scores
	UGPA score	—	T2: Higher / Lower UGPA Scores
	law school tier	—	T2: Elite / Lower Tier
COMPAS	race	White / Black	T1: White / Non-White
	gender	Male / Female	T1: Male / Female
	age	—	T2: Older (>45) / Younger (<25)
Dutch Census	gender	Male / Female	T1: Male / Female
	county of birth	—	T1: Dutch-born / Foreign-born
	citizenship	—	T1: Dutch citizens / Non-Dutch citizens
	age	—	T2: Middle (25–54) / Young & Older
	marital status	—	T2: Married / Non-married
Bank Marketing	marital status	Married / Non-married	T1: Married / Non-married
	age	Middle (25–60) / Young & Older	T1: Middle (30–55) / Young & Older
	job type	—	T2: Management / Blue-collar
	education	—	T2: Higher / Lower Education

TABLE IV

VALIDATION OF LLM-DERIVED FAIRNESS CONTEXT AGAINST ACADEMIC LITERATURE [32]–[34]. T1: STRONGLY PROTECTED ATTRIBUTES; T2: CONTEXT-DEPENDENT.

APPENDIX B

ADDITIONAL METHODOLOGICAL DETAILS

A. Aggregating Proxy-Excess Fractions Across Protected Attributes

An alternative to additive aggregation is joint conditioning: $\pi_{\text{joint}}(V_j, V_i) = \max(0, (r_{V_j V_i}^2 - r_{V_j V_i \cdot \mathbf{S}}^2) / r_{V_j V_i}^2)$, which measures the fraction attributable to the *full set* $\mathbf{S}$ simultaneously.

We choose summation for two reasons. First, joint conditioning requires $|\mathbf{S}|$ -dimensional partial associations whose reliability degrades with correlated or high-cardinality attributes.

Second, summation enforces a conservative normative commitment. Joint conditioning can allow different protected attributes to cancel each other: one attribute may induce a positive component of association while another induces a negative component, producing a small net change after conditioning on all of $\mathbf{S}$.

Summation prevents this cancellation. If either attribute individually explains part of the association, the edge receives some penalty. This is desirable when the goal is to flag any involvement of protected attributes rather than only net population-level disparity.

The cost is possible double-counting when protected attributes are correlated. Since π^Σ is used as a regularization signal rather than a causal estimand, the scale can be absorbed by λ_f , although edge rankings may still change.

B. Causal Interpretation of the Proxy-Excess Fraction

The proxy-excess fraction is a statistical quantity computed *before* the causal graph is known. The drop $r_{V_j V_i}^2 - r_{V_j V_i \cdot S_k}^2$ is causally ambiguous: it is consistent with S_k mediating the

V_j – V_i association, confounding it, or—if S_k is a collider—inducing a spurious change upon conditioning. This ambiguity is inherent to the bootstrap problem of fair causal discovery: the causal structure that would disambiguate these cases is precisely what we are trying to learn.

We argue that this ambiguity is acceptable for three reasons.

First, π functions as a *regularization penalty*, not a causal estimand: it flags edges whose pairwise association is statistically entangled with S_k , without asserting the mechanism. Edges with high π are penalized—not removed—and only to the degree that λ_f dictates, preserving a graded response that the practitioner controls.

Second, the two-phase architecture mitigates the worst failure modes. If Phase 1 retains an edge whose π value reflects confounding rather than mediation, Phase 2 re-evaluates it using the learned Markov blanket structure, where the distinction between mediation and confounding is partially resolved by the skeleton and compelled orientations.

Third, in the pure collider case—where conditioning on S_k *inflates* rather than deflates the partial association—the $\max(0, \cdot)$ operator in Eq. (3) assigns $\pi = 0$, correctly leaving the edge unpenalized. When S_k simultaneously participates in both collider and mediator/confounder structures relative to V_j – V_i , the net effect of conditioning is ambiguous and π may over- or under-penalize. Because the remaining conflation between mediation and confounding produces a *conservative* bias—penalizing S -confounded edges as if they were S -mediated—the error direction is the safer one in fairness-sensitive applications, where false negatives (undetected proxy pathways) carry greater normative cost than false positives (over-penalized edges that can be restored by reducing λ_f).

C. Sample-Size Scaling of the Fairness Penalty

The base score’s log-likelihood term grows as $O(N)$. A fairness penalty must match this rate; otherwise, as data accumulate, the statistical evidence for discriminatory edges eventually overwhelms any fixed or $O(\log N)$ counterweight, and the learned graph converges to one that faithfully encodes historical discrimination.

A natural concern is whether $O(N)$ scaling leads to the opposite failure: does the penalty eventually dominate, collapsing the graph into the empty skeleton? It does not, because the proxy-excess fraction π^Σ is itself a ratio of consistent estimators (squared correlations or mutual information and their partial counterparts) and therefore converges to a stable population quantity π_∞^Σ as N grows. The noise-floor threshold $\tau = c/(N-2)$ ensures this convergence is well-behaved by preventing division by sampling noise at finite N . Because both the log-likelihood and the penalty scale as $O(N)$, their ratio stabilizes: the effective inclusion threshold for each edge reduces to $\lambda_f \cdot \pi_\infty^\Sigma \cdot \bar{\rho}$, independent of sample size. Edges with $\pi_\infty^\Sigma \approx 0$ —those whose association is not S -mediated—face near-zero penalty at any λ_f and are included whenever the data support them. The algorithm therefore converges not to the empty graph but to a regularized graph determined by λ_f and the population proxy structure, much as ℓ_1 regularization converges to its regularized optimum rather than to the zero vector.

This does mean that $\text{Score}_{\text{PROXY}}$ is not a consistent estimator of the data-generating DAG: edges with non-negligible π_∞^Σ remain penalized regardless of sample size. This is the intended behavior—fair causal discovery requires that the learned structure not converge to one that encodes discrimination, even when that structure is statistically well-supported.

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